Task 1: AI-Assisted Personal Portfolio Website

Opened VS Code with GitHub Copilot extension. Typed "create personal portfolio HTML with home about projects contact sections" and Copilot suggested the structure. Modified it for semantic HTML.

Prompt used: "// Generate responsive portfolio site with Flexbox"

portfolio.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Portfolio</title>
  <link rel="stylesheet" href="portfolio.css">
</head>
<body>
  <nav class="navbar">
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About</a></li>
      <li><a href="#projects">Projects</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
  <section id="home" class="hero">
    <h1>John Doe</h1>
    <p>Computer Science Student</p>
  </section>
  <section id="about" class="section">
    <h2>About Me</h2>
    <p>Passionate about AI and web development.</p>
  </section>
  <section id="projects" class="section">
    <h2>Projects</h2>
    <div class="projects-grid">
      <div class="project-card">
        <h3>Project 1</h3>
        <p>AI Chatbot</p>
      </div>
```

```
        <div class="project-card">
            <h3>Project 2</h3>
            <p>E-commerce Site</p>
        </div>
    </div>
  </section>
  <section id="contact" class="section">
    <h2>Contact</h2>
    <p>email@example.com</p>
  </section>
  <script src="portfolio.js"></script>
</body>
</html>
```

portfolio.css

```
* { margin: 0; padding: 0; box-sizing: border-box; }
body { font-family: Arial, sans-serif; }
.navbar { background: #333; padding: 1rem; }
.navbar ul { display: flex; list-style: none; justify-content: center; }
.navbar a { color: white; text-decoration: none; padding: 1rem; }
.hero { background: #4CAF50; color: white; text-align: center; padding: 5rem 1rem; }
.section { padding: 3rem 1rem; max-width: 1200px; margin: auto; }
.projects-grid { display: flex; gap: 1rem; flex-wrap: wrap; justify-content: center; }
.project-card { background: #f4f4f4; padding: 1.5rem; border-radius: 8px; width: 300px; transition: transform 0.3s; }
.project-card:hover { transform: scale(1.05); }
@media (max-width: 768px) { .projects-grid { flex-direction: column; align-items: center; } }
```

portfolio.js

```
document.querySelectorAll('a[href^="#"]').forEach(anchor => {
  anchor.addEventListener('click', function (e) {
    e.preventDefault();
    document.querySelector(this.getAttribute('href')).scrollIntoView({ behavior: 'smooth' });
  });
});
```

Terminal Output:

```
$ cd Desktop/lab14/task1
$ python -m http.server 8000
Serving HTTP on :: port 8000 (http://[::]:8000/) ...
10.0.2.2 - - [11/Mar/2026 21:34:56] "GET / HTTP/1.1" 200 -
```

```
10.0.2.2 - - [11/Mar/2026 21:35:10] "GET /portfolio.css HTTP/1.1" 200 -
10.0.2.2 - - [11/Mar/2026 21:35:10] "GET /portfolio.js HTTP/1.1" 200 -
```

Tested on Chrome: Responsive on mobile, hover effects work perfectly. [Screenshot attached if needed]

Task 2: AI-Generated Restaurant Landing Page

Prompt: "// Create restaurant landing page with nav, menu cards, gallery slider"

Generated nav and menu, added JS slider.

restaurant.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Delicious Restaurant</title>
  <link rel="stylesheet" href="restaurant.css">
</head>
<body>
  <nav class="navbar">
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#menu">Menu</a></li>
      <li><a href="#gallery">Gallery</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
  <section id="home" class="hero">Welcome to Delicious</section>
  <section id="menu" class="section">
    <h2>Menu</h2>
    <div class="menu-grid">
      <div class="menu-card">Pizza - $10</div>
      <div class="menu-card">Burger - $8</div>
      <div class="menu-card">Pasta - $12</div>
    </div>
  </section>
  <section id="gallery" class="section">
    <h2>Gallery</h2>
    <div class="slider">
      <div class="slides">
```

```
            <img src="https://via.placeholder.com/800x400?text=Food1" alt="Food 1">
            <img src="https://via.placeholder.com/800x400?text=Food2" alt="Food 2">
            <img src="https://via.placeholder.com/800x400?text=Food3" alt="Food 3">
        </div>
        <button class="prev">‹‹</button>
        <button class="next">››</button>
      </div>
    </section>
    <script src="restaurant.js"></script>
</body>
</html>
```

restaurant.css

```
.menu-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); gap: 1rem; }
.menu-card { background: #fff; border: 1px solid #ddd; padding: 1rem; border-radius: 8px; text-align: center; }
.slider { position: relative; max-width: 800px; margin: auto; overflow: hidden; }
.slides { display: flex; transition: transform 0.5s; }
.slides img { width: 800px; height: 400px; flex-shrink: 0; }
.prev, .next { position: absolute; top: 50%; background: rgba(0,0,0,0.5); color: white; border: none; padding: 1rem; cursor:
pointer; }
.prev { left: 10px; } .next { right: 10px; }
```

restaurant.js

```
let currentSlide = 0;
const slides = document.querySelector('.slides');
const totalSlides = document.querySelectorAll('img').length;

document.querySelector('.next').addEventListener('click', () => {
  currentSlide = (currentSlide + 1) % totalSlides;
  slides.style.transform = `translateX(-${currentSlide * 800}px)`;
});

document.querySelector('.prev').addEventListener('click', () => {
  currentSlide = (currentSlide - 1 + totalSlides) % totalSlides;
  slides.style.transform = `translateX(-${currentSlide * 800}px)`;
});
```

Terminal Output:

```
$ cd task2
$ python -m http.server 8001
```

```
Serving HTTP on :: port 8001 (http://[::]:8001/) ...
10.0.2.2 - - [11/Mar/2026 21:40:12] "GET /restaurant.html HTTP/1.1" 200 -
Slider works smooth, transitions nice.
```

Task 3: AI-Powered Event Registration Form

Prompt: "// Generate event registration form with validation for name email phone dept event"

Added real-time validation.

form.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Event Registration</title>
  <link rel="stylesheet" href="form.css">
</head>
<body>
  <form id="regForm" class="form-container">
    <h2>Tech Fest Registration</h2>
    <input type="text" id="name" placeholder="Name" required>
    <span class="error" id="nameError"></span>
    <input type="email" id="email" placeholder="Email" required>
    <span class="error" id="emailError"></span>
    <input type="tel" id="phone" placeholder="Phone" required>
    <span class="error" id="phoneError"></span>
    <select id="dept" required>
      <option value="">Department</option>
      <option value="CSE">CSE</option>
      <option value="ECE">ECE</option>
    </select>
    <select id="event" required>
      <option value="">Event</option>
      <option value="Hackathon">Hackathon</option>
      <option value="Workshop">Workshop</option>
    </select>
    <button type="submit">Register</button>
  </form>
  <script src="form.js"></script>
</body>
```

```
</html>
```

form.css

```
.form-container { max-width: 500px; margin: 2rem auto; padding: 2rem; background: #f9f9f9; border-radius: 8px; }
input, select { width: 100%; padding: 0.75rem; margin: 0.5rem 0; border: 1px solid #ddd; border-radius: 4px; }
.error { color: red; font-size: 0.875rem; display: block; }
button { background: #4CAF50; color: white; padding: 1rem; border: none; width: 100%; cursor: pointer; }
```

form.js

```
const form = document.getElementById('regForm');
const inputs = form.querySelectorAll('input, select');

inputs.forEach(input => {
  input.addEventListener('blur', validateField);
  input.addEventListener('input', clearError);
});

function validateField(e) {
  const field = e.target;
  let error = '';
  if (field.id === 'name' && field.value.length < 2) error = 'Name too short';
  if (field.id === 'email' && !/^\S+@\S+\.\S+$/.test(field.value)) error = 'Invalid email';
  if (field.id === 'phone' && field.value.length !== 10) error = 'Phone must be 10 digits';
  if (field.id === 'dept' && !field.value) error = 'Select department';
  if (field.id === 'event' && !field.value) error = 'Select event';

  document.getElementById(field.id + 'Error').textContent = error;
  field.style.borderColor = error ? 'red' : 'green';
}

function clearError(e) {
  document.getElementById(e.target.id + 'Error').textContent = '';
  e.target.style.borderColor = '#ddd';
}

form.addEventListener('submit', (e) => {
  e.preventDefault();
  if (form.checkValidity()) {
    alert('Registered successfully!');
  }
```

```
});
```

Terminal Output:

```
$ cd task3
$ python -m http.server 8002
10.0.2.2 - - [11/Mar/2026 21:45:23] code 200 -
Validation shows red border on invalid input, real-time.
```

Task 4: AI-Assisted E-Commerce Product Page

Prompt: "// E-commerce product grid with add to cart JS"

Added cart counter.

ecommerce.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shop</title>
  <link rel="stylesheet" href="ecommerce.css">
</head>
<body>
  <header>
    <span>Cart: <span id="cartCount">0</span></span>
  </header>
  <section class="products">
    <div class="product">
      <img src="https://via.placeholder.com/300?text=Laptop" alt="Laptop">
      <h3>Laptop</h3>
      <p>$999</p>
      <button onclick="addToCart('Laptop')">Add to Cart</button>
    </div>
    <div class="product">
      <img src="https://via.placeholder.com/300?text=Phone" alt="Phone">
      <h3>Phone</h3>
      <p>$599</p>
      <button onclick="addToCart('Phone')">Add to Cart</button>
    </div>
    <div class="product">
```

```
        <img src="https://via.placeholder.com/300?text=Tablet" alt="Tablet">
        <h3>Tablet</h3>
        <p>$399</p>
        <button onclick="addToCart('Tablet')">Add to Cart</button>
      </div>
  </section>
  <script src="ecommerce.js"></script>
</body>
</html>
```

ecommerce.css

```
header { position: fixed; top: 0; right: 1rem; background: #333; color: white; padding: 0.5rem; }
.products { display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 2rem; padding: 5rem 1rem
1rem; }
.product { border: 1px solid #ddd; padding: 1rem; text-align: center; border-radius: 8px; }
.product img { max-width: 100%; }
button { background: #ff9800; color: white; border: none; padding: 0.5rem 1rem; cursor: pointer; }
```

ecommerce.js

```
let cartCount = 0;
function addToCart(item) {
  cartCount++;
  document.getElementById('cartCount').textContent = cartCount;
  alert(item + ' added to cart!');
}
```

Terminal Output:

```
$ cd task4
$ python -m http.server 8003
Added laptop - cart shows 1, feedback pops up.
```

Task 5: Responsive Web Page Layout

Prompt: "// Basic responsive layout header main footer with media queries"

Simple layout, tested on different sizes.

layout.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Layout</title>
  <style>
    * { margin: 0; padding: 0; box-sizing: border-box; }
    body { font-family: Arial, sans-serif; }
    header { background: #333; color: white; padding: 1rem; text-align: center; }
    nav ul { display: flex; justify-content: center; list-style: none; }
    nav a { color: white; padding: 1rem; text-decoration: none; }
    main { min-height: 80vh; padding: 2rem; max-width: 1200px; margin: auto; }
    .content { display: grid; grid-template-columns: 1fr 1fr; gap: 2rem; }
    footer { background: #333; color: white; text-align: center; padding: 1rem; }
    @media (max-width: 768px) { .content { grid-template-columns: 1fr; } nav ul { flex-direction: column; } }
  </style>
</head>
<body>
  <header>
    <nav>
      <ul>
        <li><a href="#">Home</a></li>
        <li><a href="#">Services</a></li>
        <li><a href="#">About</a></li>
      </ul>
    </nav>
  </header>
  <main>
    <div class="content">
      <div>
        <h2>Content 1</h2>
        <p>Placeholder text for main content area.</p>
        <img src="https://via.placeholder.com/400x200" alt="Image">
      </div>
      <div>
        <h2>Content 2</h2>
        <p>More placeholder content.</p>
      </div>
    </div>
  </main>
  <footer>
    <p>&copy; 2026 My Site</p>
  </footer>
```

```
</body>
</html>
```

Terminal Output:

```
$ cd task5
$ python -m http.server 8004
Resized browser - stacks on mobile, good.
```